

CSE221: Algorithms

Graphs

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Inspiring Excellence

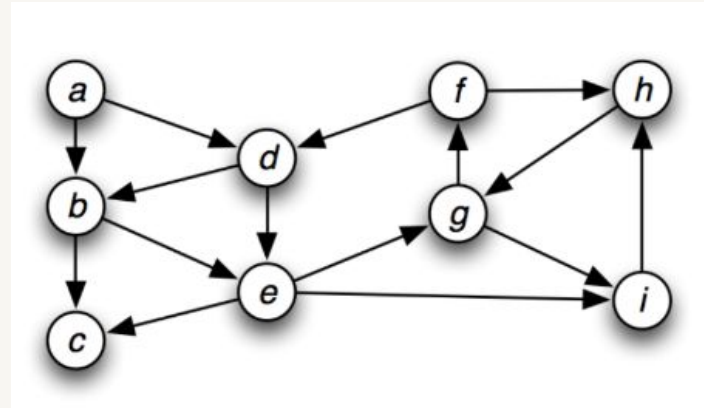
What is a Graph?

- Graph G is pair (V, E) where V is a finite set (set of vertices) and E is a finite set of pairs from V (set of edges).
- We generally denote $n = |V|$ and $m = |E|$.

Here,

$V = \{a, b, \dots, i\}$ and $n = |V| = 9$

$E = \{ab, bc, ad, \dots\}$ and $m = |E| = 15$

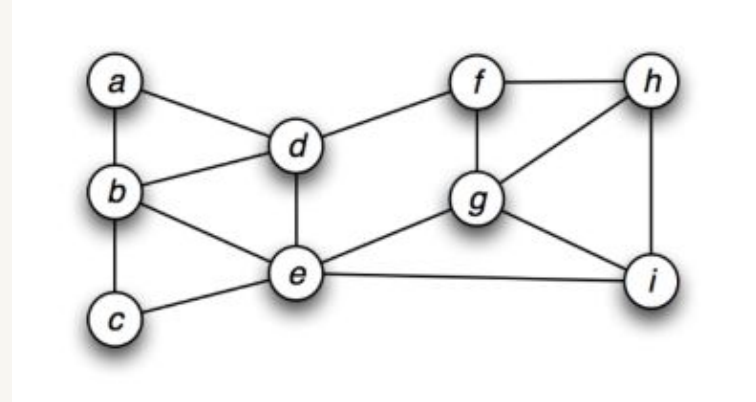
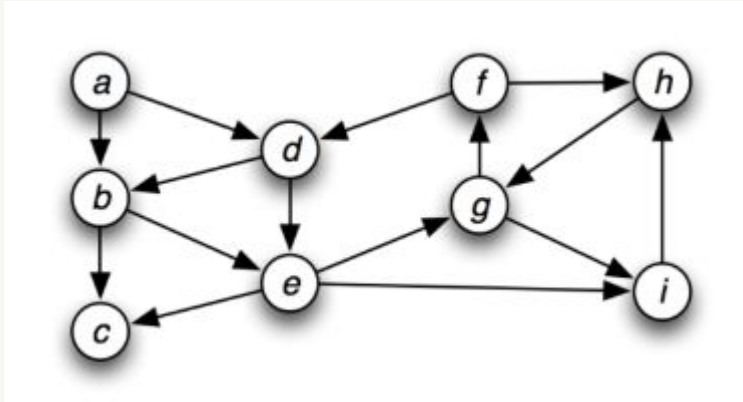


Applications of Graphs

- **Transportation Networks:** How should you design the roads and highway network in a country? What is the quickest way to go from BRAC University to Mirpur?
- **Communication Networks:** How to send network packet from your computer to your friend's computer?
- **Social Networks:** Facebook, Instagram, Twitter etc.
- **Dependency Networks:** What courses you must take before you can take CSE470?

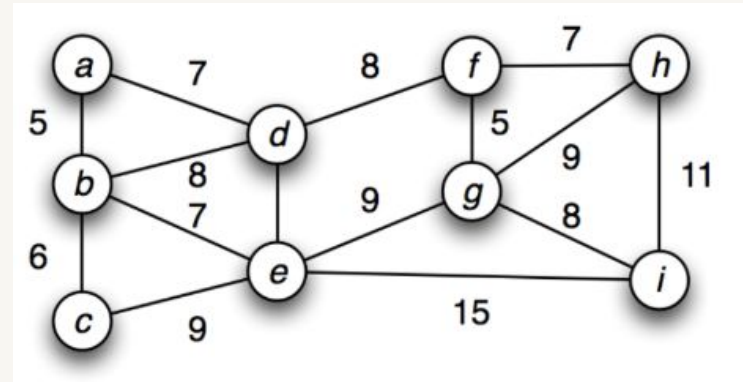
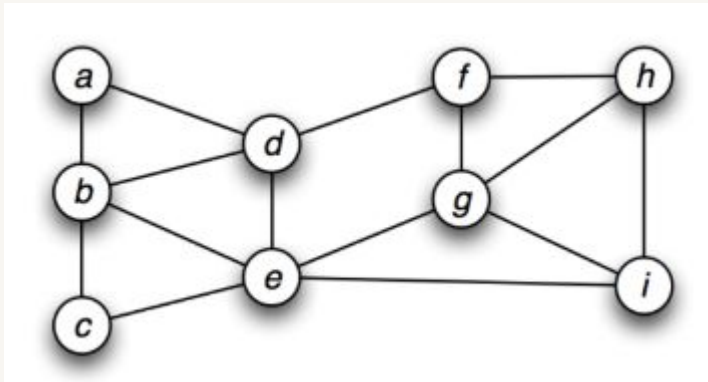
Directed vs Undirected Graphs

- Graph G can be **directed**, if E consists of ordered pairs, or **undirected**, if E consists of unordered pairs.
- If $(u, v) \in E$, then vertices u and v are adjacent.



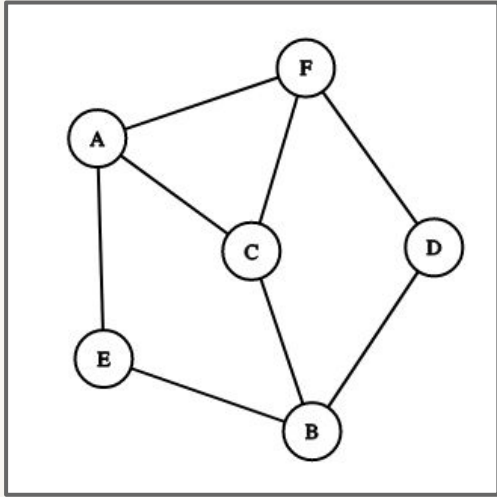
Weighted vs Unweighted Graphs

- Weights can be assigned to the edges i.e. $w_G(e)$ is a **weight of the edge $e \in E$** . Any graph that has such function assigned to it's edges is called **weighted graph**.



Basic Terminologies

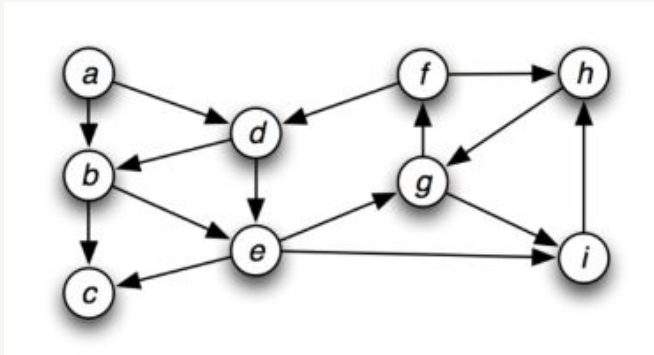
- **Edges:** Connections/Links between vertices (or nodes).



Here, AF, AC, BC etc. are edges of the graph.

Basic Terminologies

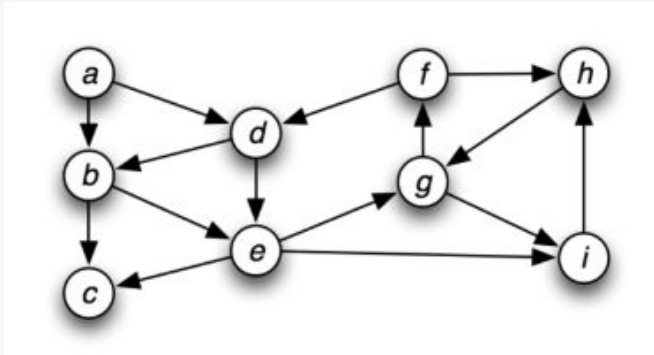
- **Incoming Edges:** Directed edges coming into a vertex.



Here, AB and DB are two incoming edges of vertex B.

Basic Terminologies

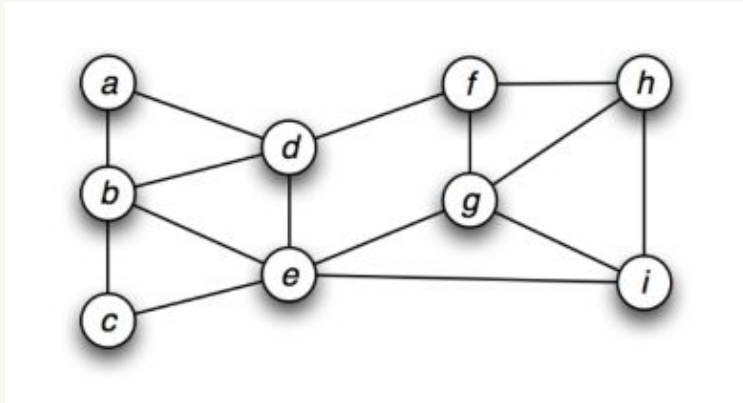
- **Outgoing Edges:** Directed edges going out of a vertex.



Here, BC and BE are two outgoing edges of vertex B.

Basic Terminologies

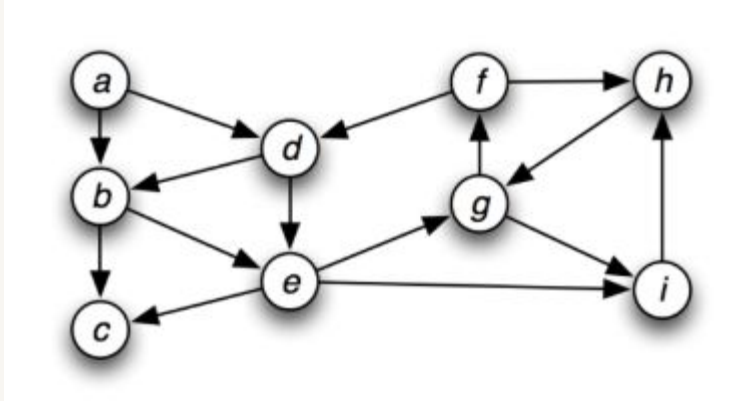
- **Degree:** For an undirected graph, **number of connected edges** to a vertex v is called the **degree of that vertex** and is denoted by $\deg(v)$.



Vertex	$\deg(v)$
a	2
b	4
c	2
d	4
e	5

Basic Terminologies

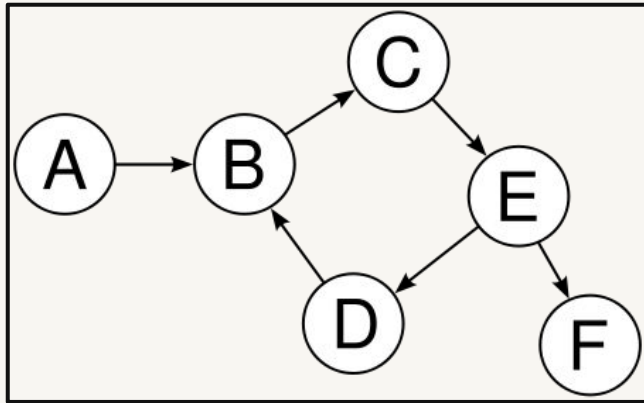
- **In Degree:** The **number of incoming edges** to a vertex v is called the **in-degree of the vertex** and is denoted by $\text{indeg}(v)$.
- **Out Degree:** The **number of outgoing edges** from a vertex v is called the **out-degree** and is denoted by $\text{outdeg}(v)$.



Vertex	$\text{indeg}(v)$	$\text{outdeg}(v)$
a	0	2
b	2	2
c	2	0
d	2	2
e	2	3

Basic Terminologies

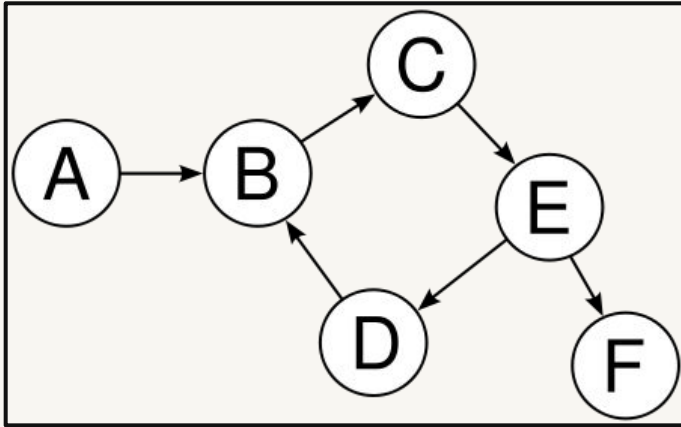
- **Source Vertex:** A vertex with in-degree zero. Generally, in a graph, at least one source vertex is present.



In the graph, vertex A is a source vertex.

Basic Terminologies

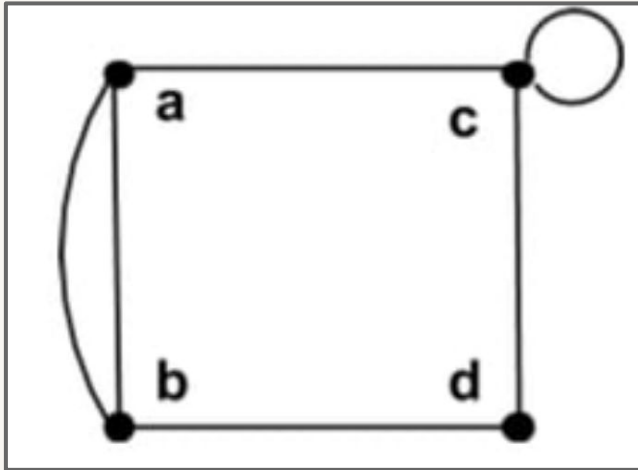
- **Sink Vertex:** A vertex with out-degree zero.



In the graph, vertex F is a sink vertex.

Basic Terminologies

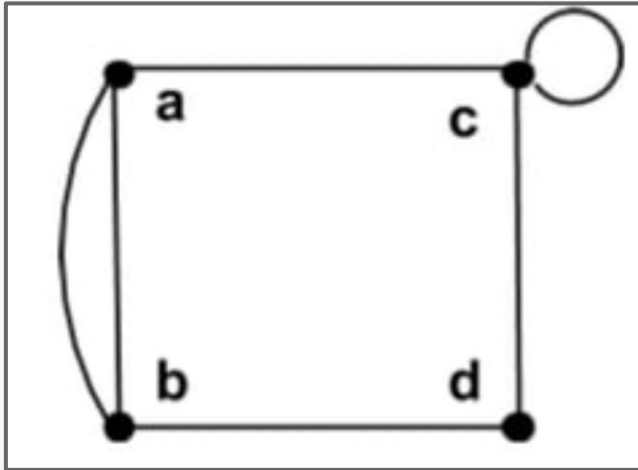
- **Parallel Edges:** Multiple edges between same pair of vertices.



In the graph, vertex a and b has two edges between each other i.e. they are parallel.

Basic Terminologies

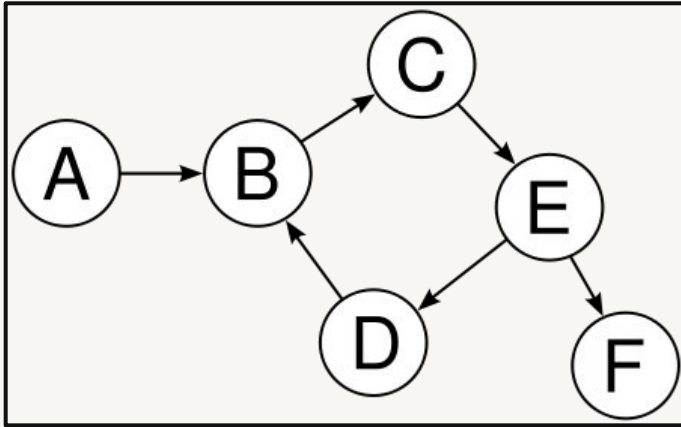
- **Self Loop:** An edge between the vertex and itself.



In the graph, vertex c has an edge which starts from c and ends at c i.e. a self loop.

Basic Terminologies

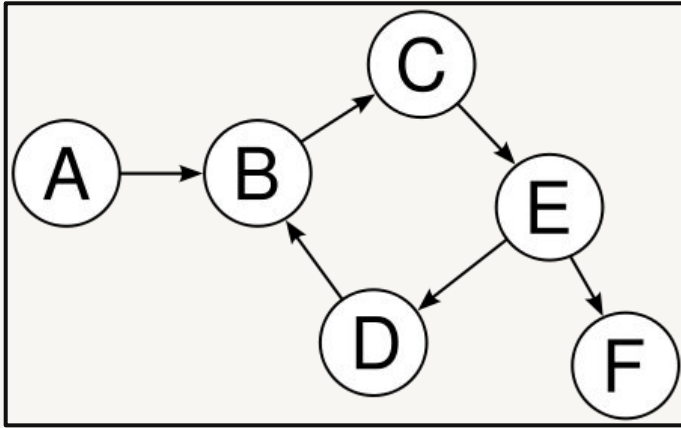
- **Path:** A sequence of vertices connected by edge(s).



In the graph, path from A to D is (A -> B -> C -> E -> D).

Basic Terminologies

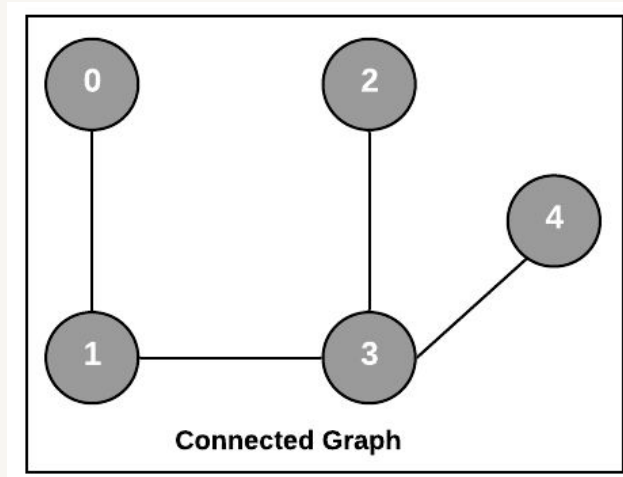
- **Cycle:** A closed path where first and last vertices are the same.



In the graph, a path exists from B to B (B -> C -> E -> D -> B) which is a cycle.

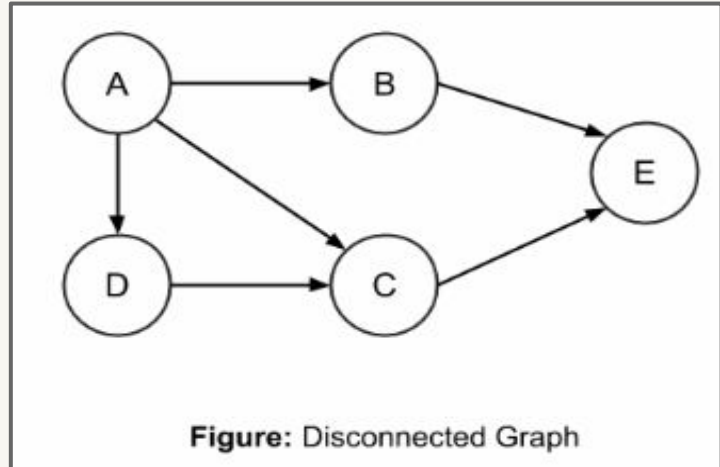
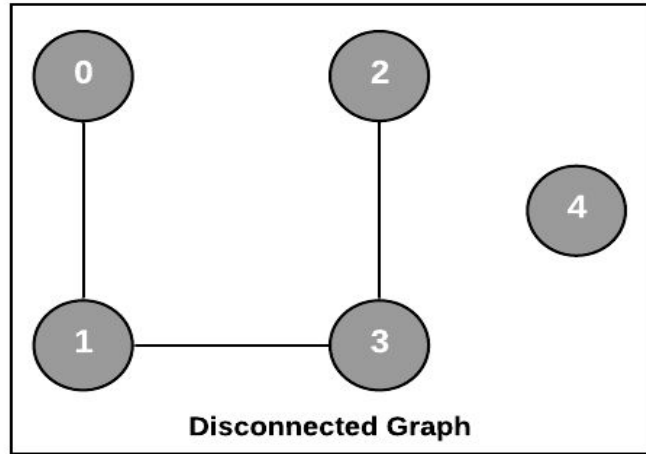
Basic Terminologies

- **Connected Graph:** A path exists between every pair of vertices.



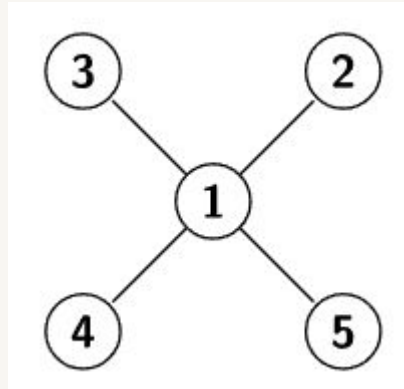
Basic Terminologies

- **Disconnected Graph:** A path is absent between any pair of vertices.



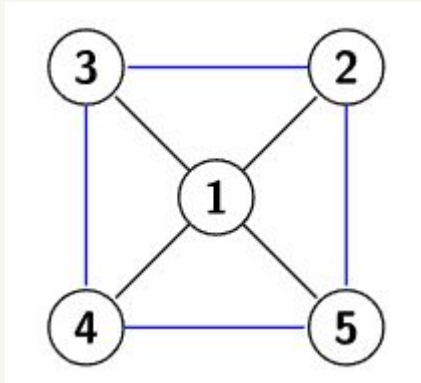
Basic Terminologies

- **Sparse Graph:** Number of edges is considerably less than maximum number of edges.



Basic Terminologies

- **Dense Graph:** Number of edges is close to the maximum number of edges.



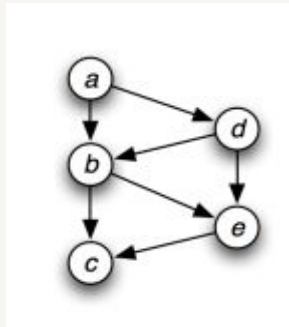
Relationship between n and m

- If G is a graph with m edges, then $\sum \deg(v) = 2m$
- If G is a directed graph with m edges, then $\sum \text{indeg}(v) = \sum \text{outdeg}(v) = m$
- If G is a simple undirected graph with n vertices and m edges, then $m \leq n(n - 1)/2$.
- If G is a simple directed graph with n vertices and m edges, then $m \leq n(n - 1)$.

Representation of Graphs

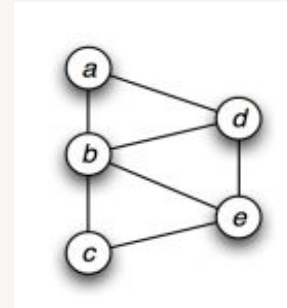
- **Adjacency Matrix:**

- A graph $G = (V, E)$ can be represented in an matrix of size $n \times n$.
- If $(v_i, v_j) \in E$, then $a_{i,j} = 1$ and 0 otherwise. In case of undirected graph, $a_{i,j} = 1$ and also $a_{j,i} = 1$.
- Memory complexity is $O(n^2)$



Directed Graph

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
<i>a</i>	0	1	0	1	0
<i>b</i>	0	0	1	0	1
<i>c</i>	0	0	0	0	0
<i>d</i>	0	1	0	0	1
<i>e</i>	0	0	1	0	0



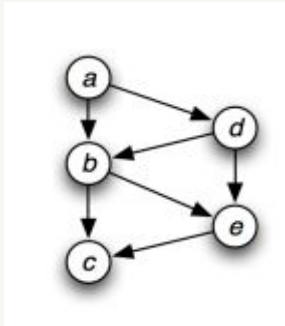
Undirected Graph

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
<i>a</i>	0	1	0	1	0
<i>b</i>	1	0	1	1	1
<i>c</i>	0	1	0	0	1
<i>d</i>	1	1	0	0	1
<i>e</i>	0	1	1	1	0

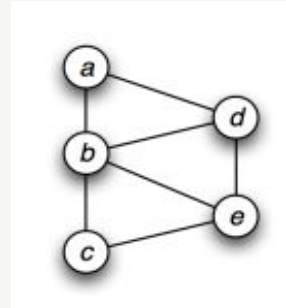
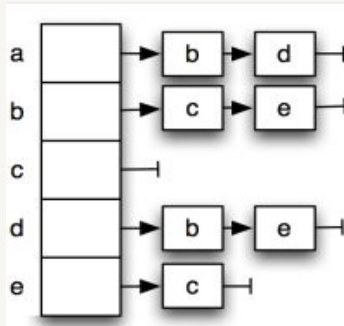
Representation of Graphs

- **Adjacency List:**

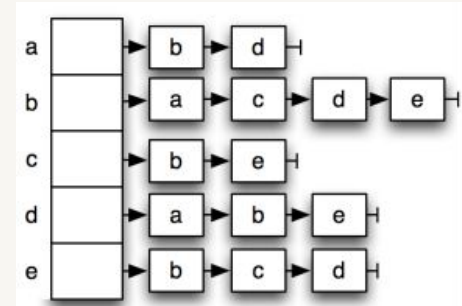
- Represents the graph by listing for each vertex v_i , it's outgoing/connected vertices in a list **$out(v_i)$** . (Can be represented using a Linked List or any other appropriate data structure)
- Memory complexity is **$O(m + n)$**



Directed Graph



Undirected Graph



References

- Mumit sir's slide -
<https://drive.google.com/file/d/1OkLO91GV4M23e-1EtvFr9yeGWgk6V91q/view>
- PNP sir's slide -
https://docs.google.com/presentation/d/1bKKT_3Mnlp0cV509k7E8hBWEKQ_YfV0cfek10LDCo6Y/edit?slide=id.g26e6291b7d8_0_183#slide=id.g26e6291b7d8_0_183